

CHANNEL LIGHTSHIP, United Kingdom

WMO: 995920

Lat: 49.900N

Lon: 2.900W

Elev: 0

StdP: 101.33

Time Zone: 0.00 (EUW)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.6	3.7	-4.1	2.7	3.8	-2.7	3.0	4.5	22.7	8.5	21.2	8.5	11.7	30	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	18.5	17.0	17.9	16.5	17.4	16.1	17.4	18.1	16.9	17.5	16.5	17.2	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.2	17.8	16.6	11.8	17.4	16.1	11.5	17.0	49.0	18.1	47.4	17.6	46.2	17.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	18.2	16.2	DB	1.4	20.2	1.6	1.6	0.2	21.4	-0.8	22.3	-1.7	23.2	-2.9	24.4	(4)
(5)			WB	-0.6	18.3	1.2	1.3	-1.5	19.3	-2.2	20.0	-2.9	20.8	-3.8	21.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	8.7	8.1	8.6	9.6	11.4	13.6	15.5	16.3	15.7	14.1	11.8	9.7	(6)	
	DBStd	3.33	2.28	2.12	1.69	1.50	1.31	1.15	1.09	0.88	1.08	1.58	1.93	2.34	(7)	
	HDD10.0	222	51	56	47	23	3	0	0	0	0	1	7	34	(8)	
	HDD18.3	2332	299	286	303	262	214	141	88	63	79	130	197	269	(9)	
	CDD10.0	932	9	3	3	12	47	109	171	195	171	129	61	23	(10)	
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.6	10.9	10.1	8.9	7.6	7.4	6.5	6.6	6.6	8.0	9.6	10.5	11.0	(14)	
(15) Precipitation	PrecAvg	839	95	74	56	54	50	46	42	57	55	92	111	107	(15)	
	PrecMax	1032	196	146	165	135	115	104	81	136	146	172	224	243	(16)	
	PrecMin	636	12	12	6	1	9	6	9	9	6	15	29	21	(17)	
	PrecStd	113	48	39	33	33	30	25	21	31	31	40	49	50	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	12.7	12.1	13.1	15.5	17.5	19.2	19.7	18.9	17.9	15.8	14.2	(19)	
		MCWB	11.6	10.8	10.3	11.3	13.2	15.5	17.4	18.1	16.8	16.4	14.3	12.6	(20)	
	2%	DB	12.6	11.7	11.2	12.2	14.1	16.3	18.1	18.6	17.9	17.1	15.1	13.7	(21)	
		MCWB	11.2	10.2	9.7	10.7	12.6	14.8	16.8	17.2	16.6	15.8	13.7	12.3	(22)	
	5%	DB	12.0	11.1	10.9	11.7	13.5	15.6	17.5	18.0	17.5	16.6	14.7	13.2	(23)	
		MCWB	10.7	9.7	9.4	10.3	12.1	14.2	16.2	16.6	16.1	15.3	13.2	11.8	(24)	
	10%	DB	11.5	10.6	10.5	11.2	12.9	15.1	17.0	17.5	17.1	16.1	14.2	12.7	(25)	
		MCWB	10.1	9.2	9.1	10.0	11.7	13.8	15.7	16.2	15.7	14.8	12.7	11.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	11.2	11.0	11.8	13.7	16.2	17.9	18.5	17.6	16.8	14.7	13.2	(27)	
		MCDB	12.7	12.3	11.5	12.6	14.7	17.1	18.6	19.4	18.0	17.3	15.5	13.8	(28)	
	2%	WB	11.5	10.5	10.2	11.1	13.0	15.2	17.1	17.6	17.0	16.1	14.1	12.6	(29)	
		MCDB	12.3	11.5	10.9	11.9	13.7	15.9	17.8	18.3	17.6	16.8	14.9	13.5	(30)	
	5%	WB	10.9	10.0	9.8	10.7	12.5	14.6	16.5	17.1	16.6	15.7	13.6	12.1	(31)	
		MCDB	11.7	10.9	10.6	11.4	13.2	15.3	17.2	17.7	17.2	16.4	14.5	13.1	(32)	
	10%	WB	10.3	9.5	9.4	10.2	12.0	14.1	16.0	16.6	16.1	15.2	13.0	11.5	(33)	
		MCDB	11.4	10.5	10.3	11.1	12.7	14.8	16.7	17.3	16.9	16.0	14.1	12.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.4	2.3	2.0	2.0	1.9	1.9	1.8	1.7	1.8	2.0	2.2	2.5	(35)	
		MCDBR	2.6	2.5	2.1	2.3	2.5	2.8	2.5	2.3	2.3	2.2	2.2	2.6	(36)	
	5% WB	MCWBR	3.2	3.1	2.3	2.1	2.1	2.3	2.2	2.2	2.2	2.4	2.8	3.4	(37)	
		MCWBR	2.6	2.5	2.1	2.2	2.4	2.6	2.4	2.2	2.2	2.1	2.2	2.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.339	0.354	0.388	0.399	0.403	0.396	0.391	0.383	0.375	0.366	0.339	0.331	(40)	
		taud	2.382	2.386	2.308	2.308	2.352	2.404	2.421	2.462	2.464	2.450	2.431	2.352	(41)	
	Ebn at Noon	Ebn at Noon	691	779	812	845	857	864	862	853	822	757	689	638	(42)	
		Edh at Noon	67	84	107	118	118	113	109	100	91	78	63	60	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.84	1.60	2.85	4.57	5.52	5.99	5.76	4.81	3.65	2.05	1.07	0.69	(44)	
	RadStd	RadStd	0.07	0.19	0.33	0.38	0.35	0.59	0.34	0.37	0.24	0.14	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page